Architecture of Modern PDF Systems

PDF (Portable Document Format) is an international standard (ISO 32000) for representing documents independent of application software, hardware, and operating systems.

Key concepts include:
1. Low-level COS (Certificate Object Stream) object hierarchy.
2. High-level Page Description (PD) Model.
3. Content Streams containing postscript-like rendering operators.
4. Detached digital signatures using PKCS#7 / CMS standards.

Section 2: Digital Signatures and Security

Digital signatures provide authenticity, data integrity, and non-repudiation. PDF files allow incremental updates where signatures are appended without invalidating preceding byte content.